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LETTER TO THE EDITOR OPEN ACCESS

# Comment on "Variation in Repeated Handgrip Strength Testing Indicates Submaximal Force Production in Patients With Myalgic Encephalomyelitis/Chronic Fatigue Syndrome"

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Dear Editor,

We have read the recent article titled "Variation in Repeated Handgrip Strength Testing Indicates Submaximal Force Production in Patients With Myalgic Encephalomyelitis/Chronic Fatigue Syndrome" by Popkirov with great interest [1]. The author proposes that increased variability in handgrip strength (HGS) reflects higher-order motor control disturbances in individuals with ME/CFS. While we appreciate this effort, several methodological issues regarding the application of the coefficient of variation (CV) for this purpose require clarification.

First, the rationale for using the CV is difficult to reconcile with the cited literature. While the author does not imply intentional underperformance, the interpretation assumes that increased CV reflects a meaningful abnormality in force regulation. Yet the seminal review by Shechtman [2] concluded that "the CV is not an appropriate method for determining whether an effort is sincere," identifying limitations directly relevant here, including inflation of CV in weaker or injured individuals. A subsequent study by the same author, also cited by Popkirov, reiterated that CV should not be used for evaluating grip-force abnormalities due to mathematical inconsistency, limited specificity, and vulnerability to misclassification [3]. Although originally framed within sincerity-of-effort testing, these methodological concerns remain pertinent. Moreover, Wachter et al. [4] tested healthy volunteers under full-effort, reduced-effort, and simulated-injury conditions. Whereas other modalities distinguished these states, CV did not differ significantly, further questioning its interpretability in HGS testing. These caveats warrant consideration.

Second, CV calculated from rapid, repeated HGS trials is inherently confounded by fatigability. Even healthy individuals show measurable decline when repetitions are closely spaced, and ME/CFS patients exhibit a steeper decline correlated with disease severity [5]. Under these conditions, elevated CV is at least partly attributable to fatigability rather than altered motor control. Although the author excluded participants with pronounced fatigability, residual decline persists and could inflate CV (Figure 1). Also, the fatigability-based exclusion criterion removes subjects with the largest force decline, arguably representing more severe disease [5]. Such selective attrition distorts variance estimates and may obscure true group differences. Instead, more robust statistical modelling approaches would be preferable.

Third, the measurement protocol used differs substantially from those in which CV was originally proposed. Earlier studies conducted either few discrete trials with rest or analysed single-grip force–time curves – conditions unlikely to induce fatigability [2]. These findings cannot be directly transferred to a 10-repetition protocol, further limiting interpretability.

Finally, we were unable to reproduce the reported participant numbers using the publicly available dataset, despite applying the stated exclusion criteria.

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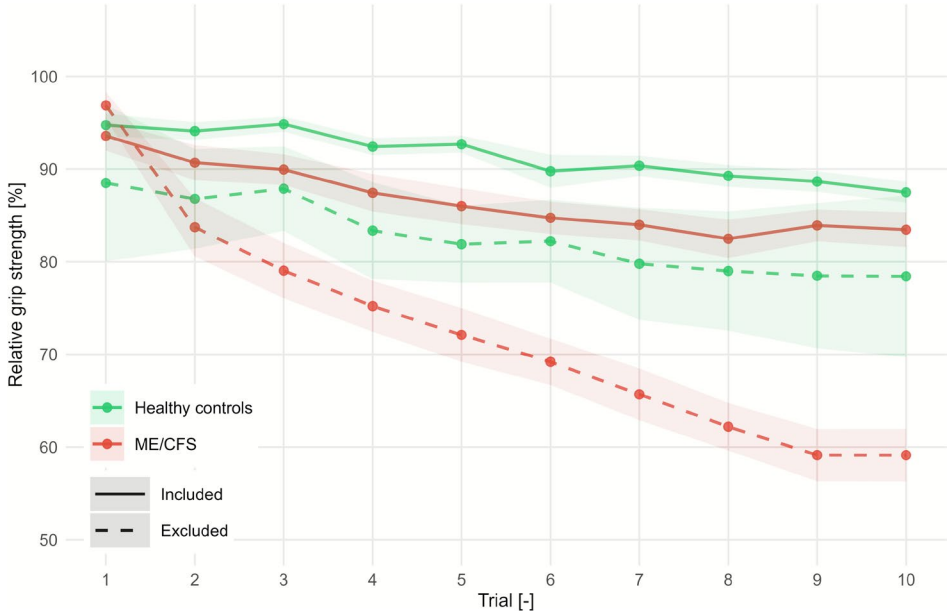



**FIGURE 1** | Relative handgrip strength (mean ± 95% CI) series as analysed by Popkirov in [1]. Solid lines represent participants included in the analysis. Healthy controls are shown in green, and individuals with ME/CFS in red.

Taken in context, the analytical framework rests on insufficiently supported methodological premises and risks producing inaccurate conclusions. In ME/CFS, interpreting variability in force output as indicative of higher-order motor processes is particularly vulnerable to misinterpretation. Such inferences should be drawn cautiously in the absence of adequately validated methodology.

We respectfully suggest that the analytical approach be revisited, that the conclusions be reconsidered or more cautiously framed, and that the methodological limitations be explicitly acknowledged.

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### Author Contributions

**Felix Steinkirchner:** methodology, visualization, writing – review and editing, data curation, formal analysis. **Viktoria Kimmerling:** methodology, writing – review and editing. **Christina Kaufmann:** visualization, writing – review and editing. **Elisabeth Schieffer:** methodology. **Michael Gruber:** conceptualization, methodology, visualization, project administration. **Alexander Dejaco:** writing – original draft, writing – review and editing, formal analysis, resources, supervision.

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### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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